Q1.

Some pupils predicted that water will evaporate faster if the surrounding air temperature is higher.

To investigate their prediction they placed some water in containers in two different rooms.

(a) Give **two** factors they should keep the same to make their investigation fair.

1. .....

1 mark

2. .....

1 mark

(b) They recorded the mass of the water and the container in room 1 and room 2 every day for 5 days.

The table below shows their results.

time (days)	**mass of water and container (g)**	
	room 1	**room 2**
0	100	100
1	92	85
2	80	72
3	72	54
4	60	45
5	46	30

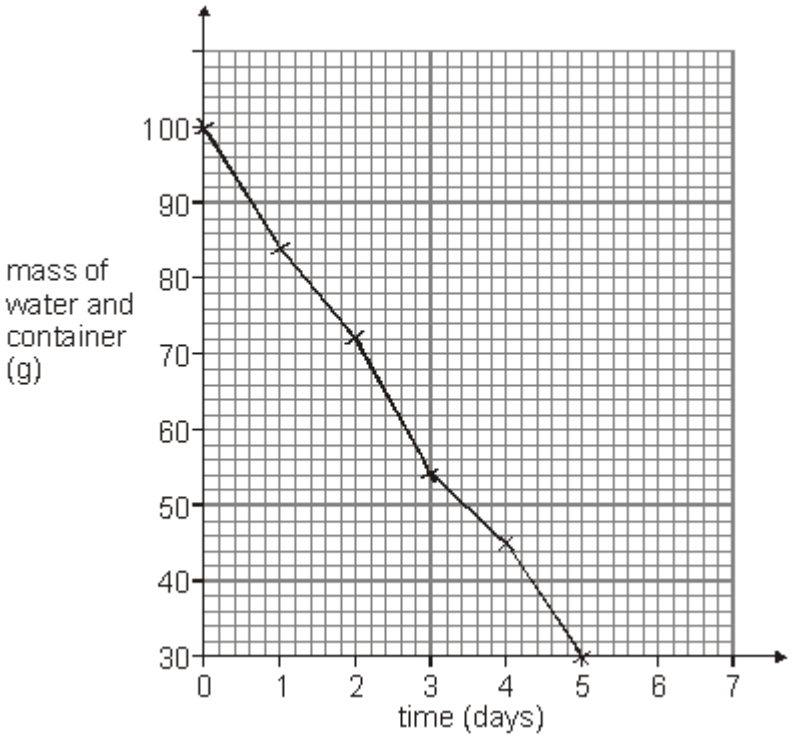
The data shown in their table is **not** sufficient to test their prediction.
Explain why.

.....

.....

1 mark

They plotted their data for room 2 and attempted to draw a line of best fit.



(c) Describe the mistake they made in drawing the line of best fit.

.....

.....

1 mark

(d) Using the data in the table plot the points for room 1.

1 mark

(e) Draw a line of best fit of the points you have drawn.

1 mark

(f) In which room did the water evaporate more quickly?
Tick **one** box.

room 1 ☐ room 2 ☐

Use their data to explain your answer.

.....

.....

1 mark
maximum 7 marks

Q1.

(a) any **two** from

- initial temperature of water
- exclusion of draughts
  accept 'same amount of draughts'
- volume **or** mass of water
  accept 'amount of water'
- surface area (of water)
  *accept 'shape **or** size **or** material of container'*
- humidity
  accept 'the time it is left for'
  'temperature' is insufficient
  do not accept 'size of room'

2 (L5)

(b)
- they have not collected data about temperature
  accept 'they did not measure temperature'
- no data recorded about temperature
  accept 'no information about temperature'
  accept 'no temperature'

1 (L5)

(c) the line is not straight (converse)
the line should be straight
accept 'the line is zig-zag'
accept 'they joined up all the points'
accept 'they didn't go between the points'
accept 'the line does not need to go through all the points'

1 (L6)

(d) 6 points correctly plotted to the nearest half square

1 (L6)

(e) a straight line of best fit between some of the points

1 (L6)

(f)

☐ ☑ room 2

and

- the points for room 2 are lower than room 1
- the container in room 2 reaches 30g in 5 days, room 1 reaches 45g
  accept 'there is less water in the container in room 2 than in

room 1'

- more water evaporated in the same time

  accept 'amount of water dropped more in the first three days'
  accept 'the line for room 2 is steeper'

- the volume goes down more quickly

  accept 'the numbers go down more quickly'
  accept 'the mass goes down more quickly'

1 (L6)

[7]